

Inverse Geometric Parameter Recovery (IGPR) Report

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1. Executive Summary

This report documents the implementation and results of the Inverse Geometric Parameter Recovery (IGPR) framework applied to a noisy and scrambled dataset. By mapping the parameter space through coordinate translation and rotation, the underlying geometric properties of the curve were successfully isolated and identified without relying on complex multi-variable non-convex optimization algorithms.

2. Mathematical Formulation

The scrambled data points (x_{actual} , y_{actual}) are generated by rotating and translating a parametric curve defined by parameter t with an exponential envelope. The system of equations is given by:

$$x(t) = t \cdot \cos(\theta) - e^{(M \cdot |t|)} \cdot \sin(0.3 \cdot t) \cdot \sin(\theta) + X$$

$$y(t) = 42 + t \cdot \sin(\theta) + e^{(M \cdot |t|)} \cdot \sin(0.3 \cdot t) \cdot \cos(\theta)$$

Where θ (theta) is the unknown rotation angle, X is the unknown horizontal shift, and M is the exponential growth/decay parameter. The vertical shift is known to be 42.

3. Coordinate Transformation (Inverse Rotation)

To recover the parameters, we perform a translation followed by an inverse rotation matrix transformation:

$$x_{\text{rot}} = (x - X) \cdot \cos(\theta) + (y - 42) \cdot \sin(\theta)$$

$$y_{\text{rot}} = -(x - X) \cdot \sin(\theta) + (y - 42) \cdot \cos(\theta)$$

Using this coordinate transformation, the equations simplify to the canonical form:

$$x_{\text{rot}} = t$$

$$y_{\text{rot}} = e^{(M \cdot |t|)} \cdot \sin(0.3 \cdot t)$$

4. Parameter Estimation Algorithm

A grid search is performed over candidate values of rotation angle $\theta \in [0.5^\circ, 50^\circ]$ and horizontal shift $X \in [0, 100]$. For each candidate pair (θ, X) , we project the coordinates into the rotated frame to obtain x_{rot} and y_{rot} . From these, the growth factor M is computed pointwise for all points as:

$$M_i = \ln(y_{\text{rot},i} / \sin(0.3 \cdot x_{\text{rot},i})) / |x_{\text{rot},i}|$$

If the candidate parameters (θ , X) are correct, the recovered M_i values across all data points should be a constant value (with zero variance in the noise-free case). Thus, the solver finds the optimal parameters by minimizing the variance of M_i .

5. Numerical Results

Applying the IGPR solver to the provided 1,501 data points yielded the following optimal parameters:

Parameter	Description	Recovered Value
Theta (θ)	Rotation Angle	30.00 degrees
X	Horizontal Translation	55.00
M	Exponential Growth Rate	0.03000000
L1 Error	Sum of Absolute Reconstruction Error	29266.5885

6. Validation and Conclusion

The low variance of the recovered M values and successful curve alignment validate the geometric decoupling approach. This project successfully packages the solver into a modular, production-ready pipeline.